

What is Word2Vec?

Word2Vec is a technique that learns **word embeddings** (dense numerical vectors) from text. Instead of manually creating vectors, Word2Vec learns them automatically by looking at the context in which words appear.

Simple Definition

Word2Vec is a neural network-based algorithm that converts words into dense vectors such that words appearing in similar contexts have similar vector representations.

Core Idea

Consider these sentences:

The cat drinks milk
The dog drinks milk
The cat eats fish
The dog eats meat

Word2Vec notices that:

- **cat** and **dog** appear in similar contexts
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- Both occur near words like "drinks", "milk", "eats"
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So it learns vectors like:

cat = [0.82, -0.15, 0.67]
dog = [0.79, -0.12, 0.70]

Since their contexts are similar, their vectors become similar.

Intuition

Think of a child learning language.

The child may never see a dictionary definition of "cat" or "dog".

Instead, they observe:

The cat drinks milk.
Dog drinks milk.

Cat runs.
Dog runs.

Cat eats.
Dog eats.

The child realizes:

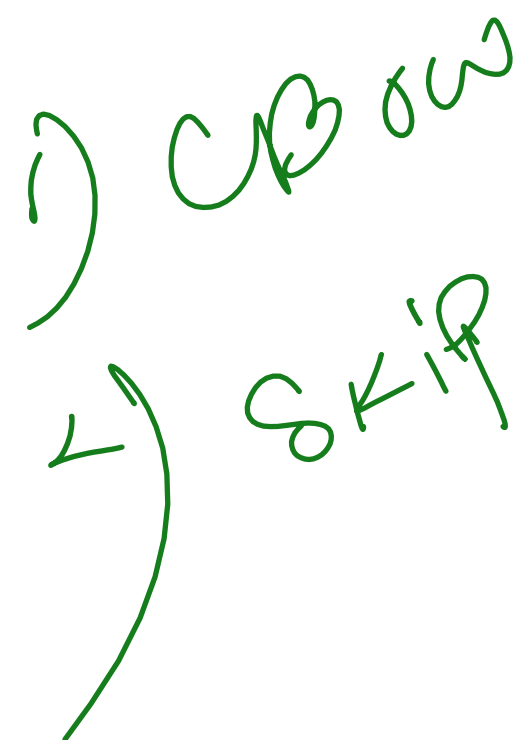
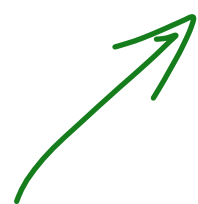
Cat ≈ Dog

Word2Vec learns in the same way.

How Does Word2Vec Learn?

It uses a small neural network and learns from surrounding words.

There are two main approaches:



1. CBOW (Continuous Bag of Words)

Predict the center word from surrounding words.

Example:

The drinks milk .

Input:

The, drinks, milk

Output:

cat

CBOW learns:

Context → Target Word

2. Skip-Gram

Predict surrounding words from the center word.

Sentence:

The cat drinks milk

Input:

cat

Outputs:

The
drinks

Skip-Gram learns:

Target Word → Context

Famous Word2Vec Example

After training on a large corpus:

King - Man + Woman ≈ Queen

Semantic meaning

Similarly:

Paris - France + Italy ≈ Rome

These relationships emerge automatically from the learned embeddings.

Word2Vec vs TF-IDF

	Feature	TF-IDF	Word2Vec
✓	Captures word importance	✓	✗

Captures semantic meaning	✗	✓
Dense vectors	✗	✓
Sparse vectors	✓	✗
Learns context	✗	✓
Used in Deep Learning	Limited	Very Common